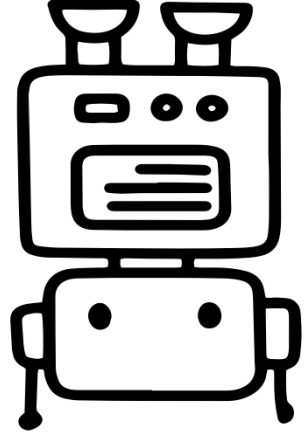


FINAL WORK

Web application development in the .NET environment

Student:

Emanuela Vidojević SRTd 01/23

SEPTEMBER 2026.

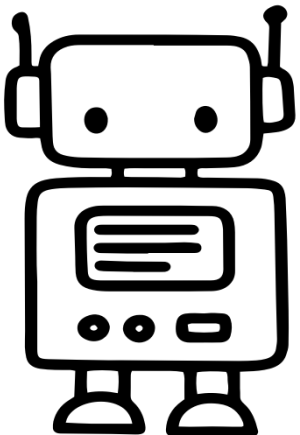
PROJECT OVERVIEW

PURPOSE AND OBJECTIVE:

- Centralized exchange materials in higher education institutions.
- Digitization and optimization academic processes.

ORGANIZATION AND FUNCTION:

- Hierarchical storage of content by subjects.
- Evaluation of materials and notifications in real time.



TARGET USER GROUP



Students

- Access and search of materials by subjects
- Automatic notifications and content monitoring.



Professors

- Management and publishing of digital resources.
- Monitoring student engagement.

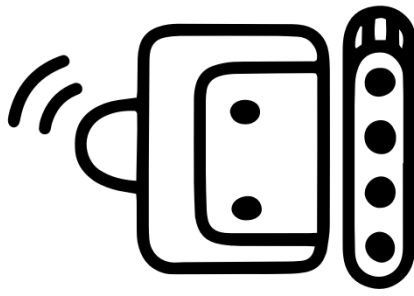


Administrators

- Managing user accounts and roles.
- Hierarchical structure configuration.

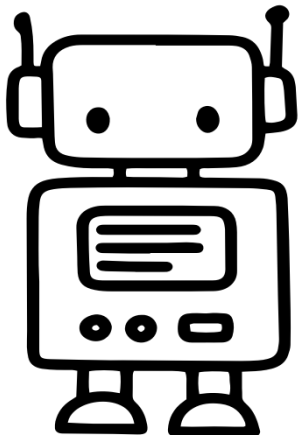
MAIN CHARACTERISTICS PROJECT

- Access Management: Secure JWT authentication and RBAC authorization.
- Content organization: Multimedia support and hierarchical structure by subjects.
- Notification system: Automatic notifications in real time.



Technology Selection

System layer	Tehnology	Description / Intended Use
Frontend	Vue.js & Bootstrap	A progressive framework and library for a fully responsive UI
Backend	.NET Core Web API (C#)	Robust, high-performance and scalable server logic
Database	MySQL & EF Core	Relational data store with ORM object mapping



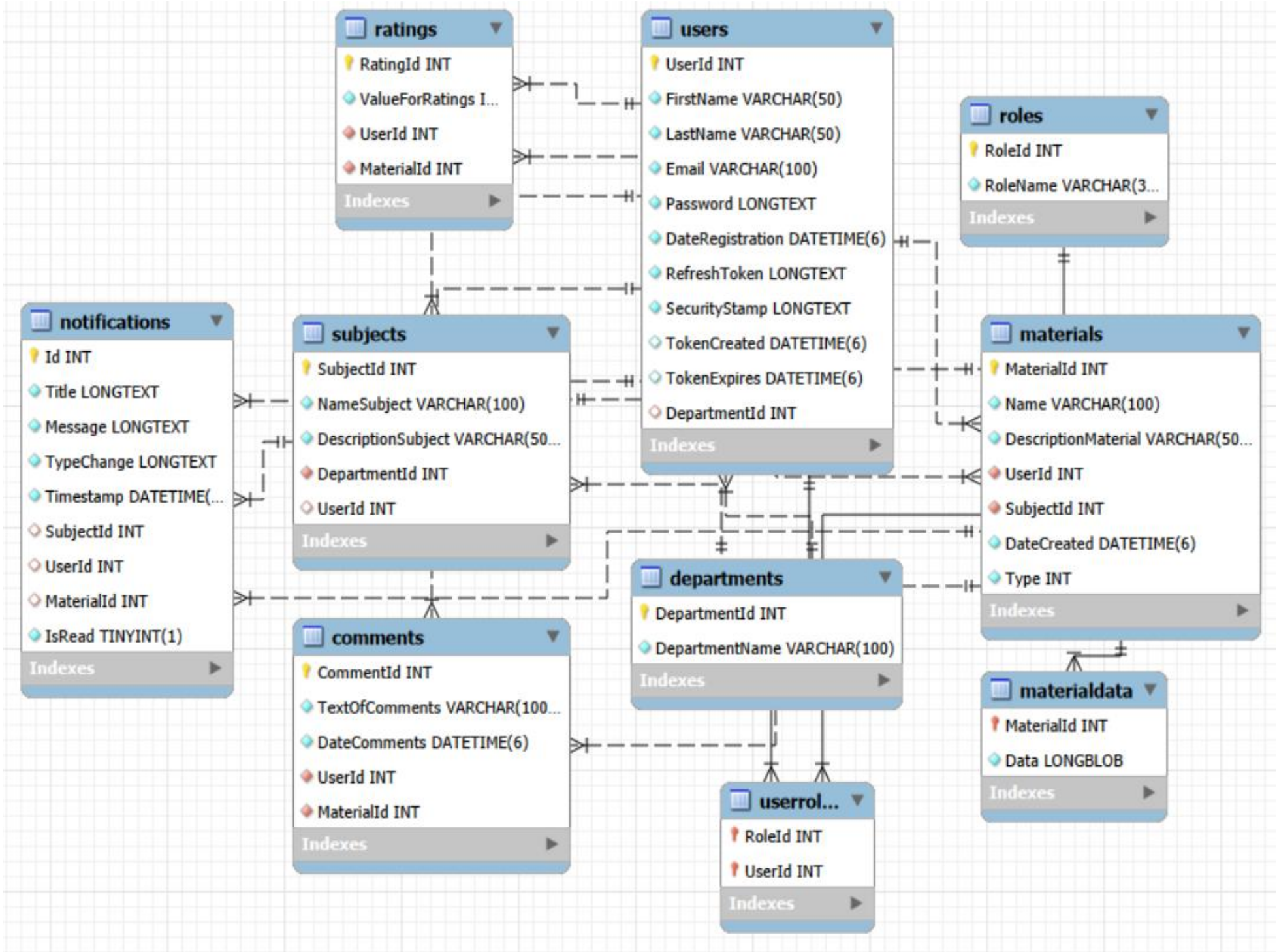
DATABASE ARCHITECTURE

•Structure and Entities: A linked relational model that spans users (Users), roles (Roles), departments (Departments), subjects (Subjects) and the materials themselves (Materials).

•Content management: Separate metadata (MaterialData) with integrated features for ratings, comments and notifications

•Security and data protection: Using secure mechanisms to hashing passwords, ensuring that sensitive user data never are not visible in clear text, with support for security keys (SecurityStamp) and session tokens.

DATABASE SCHEMA

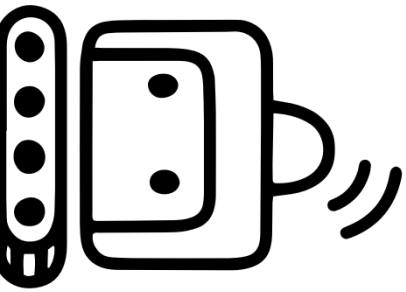


SAFETY PROTOCOLS

•Password Protection: Credential hashing and secure storage in to the base.

• API Protection: Prevention of SQL Injection and XSS attacks validation.

• Session management: Access control via Refresh Mechanism Token.



EduShare platform

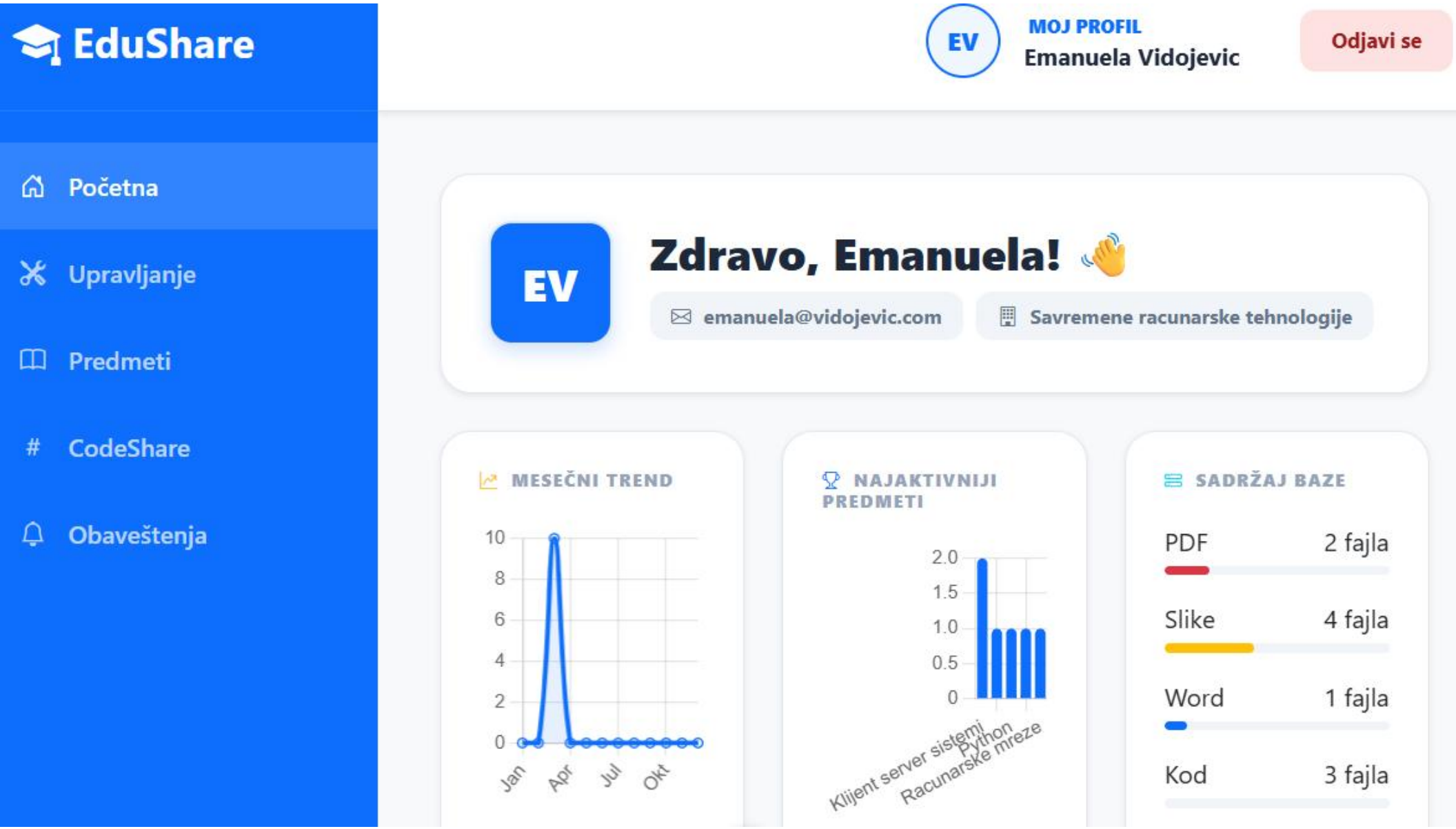
•Overview dashboard: A central place to display key statistics, monthly activity trends and structure of the material base.

• Academic Analytics: Insights into the most active subjects and statistics files by format (PDF, images, Word, code) in real time.

•Intuitive navigation: A side menu that provides quick access homepage, content management, subjects, CodeShare and notifications.



EduShare platform



FUTURE DEVELOPMENT PLAN AND CONCLUSION

Development direction	Description
Dark Mode	Implementation of dark theme for visual comfort.
AI Asistent	Smart search and assistance for students.
Mobile App	Development of a native mobile application.

CONCLUSION

- Comprehensive platform: Connects academic structure, resources and interactive cooperation in one place...
- Reliability and security: Stable database and advanced protection data guarantee safe operation.
- The ultimate experience: An intuitive interface that makes knowledge sharing possible simple and efficient.

АКАДЕМИЈА ТЕХНИЧКО-ВАСПИТАЧКИХ СТРУКОВНИХ СТУДИЈА
ОДСЕК НИШ

FINAL THESIS - WEB APPLICATION DEVELOPMENT IN .NET ENVIRONMENT

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